SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Generation and Classification of Continuous-Time and Discrete-Time Signals	
Expt / Prog No:	1
Name of the Expt / Prog	**To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.**
Date of the Expt	
CO	CO1
GITHUB Link of the Expt	

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Generate and visualize continuous-time and discrete-time elementary signals.	Signal Generation & MATLAB Implementation	Generates all required CT and DT signals accurately using suitable MATLAB commands and parameters; plots are correctly scaled and labelled.	Generates the required signals and plots correctly with minor coding, parameter, or labelling errors.	Generates basic signals but requires assistance or has noticeable errors in code or plots.	Unable to generate the required signals or produce meaningful plots.	25
Classify signals as periodic/aperiodic and identify their waveform characteristics.	Signal Classification & Analysis	Correctly classifies signals using mathematical conditions and waveform evidence; determines period where applicable and explains the reasoning clearly.	Classifies most signals correctly with minor errors or limited justification.	Provides basic classifications but requires guidance or gives incomplete justification.	Unable to classify signals correctly or justify the classification.	25
Determine even and odd components and verify reconstruction of the original signal.	Even–Odd Decomposition & Reconstruction	Correctly computes even and odd components, verifies symmetry/anti-symmetry, and reconstructs the original signal with negligible numerical error.	Obtains components and reconstruction correctly with minor computational or interpretation errors.	Performs decomposition with assistance; verification or reconstruction is incomplete or contains noticeable errors.	Unable to determine the components or verify reconstruction correctly.	30

Compare simulated outputs with theoretical expectations and communicate results.	Output Verification, Interpretation & Documentation	Accurately verifies waveforms against theory, records observations, and provides a technically sound conclusion with clear explanations.	Provides correct output comparison and conclusion with minor gaps in observations or interpretation.	Provides basic comparison and conclusion but requires guidance or omits important observations.	Unable to validate results or provide a meaningful conclusion; documentation is incomplete.	20
TOTAL						**100**

Objective / Assessment Criterion	**Level**	**Max Weightage (10)**	**Mark Obtained**	**Mark (100)**
Generate and visualize continuous-time and discrete-time elementary signals.		**2.5**		
Classify signals as periodic/aperiodic and identify their waveform characteristics.		**2.5**		
Determine even and odd components and verify reconstruction of the original signal.		**3**		
Compare simulated outputs with theoretical expectations and communicate results.		**2**		
		Total		
		Course Faculty Sign with date		

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Generation and Classification of Continuous-Time and Discrete-Time Signals	
Expt / Prog No:	2
Name of the Expt / Prog	**To generate and visualize basic discrete-time signals and analyze their amplitude and time-index characteristics.**
Date of the Expt	
CO	CO1
GITHUB Link of the Expt	

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Generate and visualize continuous-time and discrete-time elementary signals.	Signal Generation & MATLAB Implementation	Generates all required CT and DT signals accurately using suitable MATLAB commands and parameters; plots are correctly scaled and labelled.	Generates the required signals and plots correctly with minor coding, parameter, or labelling errors.	Generates basic signals but requires assistance or has noticeable errors in code or plots.	Unable to generate the required signals or produce meaningful plots.	25
Classify signals as periodic/aperiodic and identify their waveform characteristics.	Signal Classification & Analysis	Correctly classifies signals using mathematical conditions and waveform evidence; determines period where applicable and explains the reasoning clearly.	Classifies most signals correctly with minor errors or limited justification.	Provides basic classifications but requires guidance or gives incomplete justification.	Unable to classify signals correctly or justify the classification.	25
Determine even and odd components and verify reconstruction of the original signal.	Even–Odd Decomposition & Reconstruction	Correctly computes even and odd components, verifies symmetry/anti-symmetry, and reconstructs the original signal with negligible numerical error.	Obtains components and reconstruction correctly with minor computational or interpretation errors.	Performs decomposition with assistance; verification or reconstruction is incomplete or contains noticeable errors.	Unable to determine the components or verify reconstruction correctly.	30

Compare simulated outputs with theoretical expectations and communicate results.	Output Verification, Interpretation & Documentation	Accurately verifies waveforms against theory, records observations, and provides a technically sound conclusion with clear explanations.	Provides correct output comparison and conclusion with minor gaps in observations or interpretation.	Provides basic comparison and conclusion but requires guidance or omits important observations.	Unable to validate results or provide a meaningful conclusion; documentation is incomplete.	20
TOTAL						**100**

Objective / Assessment Criterion	**Level**	**Max Weightage (10)**	**Mark Obtained**	**Mark (100)**
Generate and visualize continuous-time and discrete-time elementary signals.		**2.5**		
Classify signals as periodic/aperiodic and identify their waveform characteristics.		**2.5**		
Determine even and odd components and verify reconstruction of the original signal.		**3**		
Compare simulated outputs with theoretical expectations and communicate results.		**2**		
		Total		
Course Faculty Sign with date				

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Signal Operations and Transformations	
Expt / Prog No:	3
Name of the Expt / Prog	**To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.**
Date of the Expt	
CO	CO1
GITHUB Link of the Expt	

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Implement amplitude/time scaling, shifting, reversal and sequence operations	Signal Operation & MATLAB Implementation	Implements all required signal operations accurately and independently using appropriate MATLAB commands and parameters.	Implements operations correctly with minor coding or parameter errors.	Implements basic operations but requires considerable assistance or has noticeable errors.	Unable to implement the required signal operations correctly.	25
Analyze changes produced by signal transformations	Transformation Analysis & Visualization	Clearly visualizes and accurately explains changes in amplitude, time scale, position, orientation and sequence values.	Correctly identifies major transformation effects with minor interpretation gaps.	Provides basic plots and limited interpretation of transformation effects.	Unable to interpret or explain the effect of transformations.	25
Compare transformed signals with theoretical results	Output Verification & Theoretical Comparison	Accurately verifies all transformations mathematically and numerically with negligible deviation.	Provides correct comparison with minor errors or small deviations.	Performs basic comparison but validation is incomplete or contains noticeable errors.	Unable to compare simulation output with theoretical results.	30
Interpret results and document the experiment	Engineering Interpretation, Record & Viva	Provides technically sound conclusions, complete documentation and clear viva responses independently.	Provides proper documentation and reasonable conclusions with minor gaps.	Documentation and conclusions are adequate but require guidance.	Incomplete documentation and limited understanding of results.	20

Total						100

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Implement amplitude/time scaling, shifting, reversal and sequence operations		2.5		
Analyze changes produced by signal transformations		2.5		
Compare transformed signals with theoretical results		3		
Interpret results and document the experiment		2		
Total				
Course Faculty Sign with date				

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Signal Operations and Transformations	
Expt / Prog No:	4
Name of the Expt / Prog	**To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics**
Date of the Expt	
CO	CO1
GITHUB Link of the Expt	

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Implement amplitude/time scaling, shifting, reversal and sequence operations	Signal Operation & MATLAB Implementation	Implements all required signal operations accurately and independently using appropriate MATLAB commands and parameters.	Implements operations correctly with minor coding or parameter errors.	Implements basic operations but requires considerable assistance or has noticeable errors.	Unable to implement the required signal operations correctly.	25
Analyze changes produced by signal transformations	Transformation Analysis & Visualization	Clearly visualizes and accurately explains changes in amplitude, time scale, position, orientation and sequence values.	Correctly identifies major transformation effects with minor interpretation gaps.	Provides basic plots and limited interpretation of transformation effects.	Unable to interpret or explain the effect of transformations.	25
Compare transformed signals with theoretical results	Output Verification & Theoretical Comparison	Accurately verifies all transformations mathematically and numerically with negligible deviation.	Provides correct comparison with minor errors or small deviations.	Performs basic comparison but validation is incomplete or contains noticeable errors.	Unable to compare simulation output with theoretical results.	30
Interpret results and document the experiment	Engineering Interpretation, Record & Viva	Provides technically sound conclusions, complete documentation and clear viva responses independently.	Provides proper documentation and reasonable conclusions with minor gaps.	Documentation and conclusions are adequate but require guidance.	Incomplete documentation and limited understanding of results.	20
Total						100

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Implement amplitude/time scaling, shifting, reversal and sequence operations		2.5		
Analyze changes produced by signal transformations		2.5		
Compare transformed signals with theoretical results		3		
Interpret results and document the experiment		2		
		Total		
		Course Faculty Sign with date		

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Energy and Power Analysis of Signals	
Expt / Prog No:	5
Name of the Expt / Prog	To calculate the energy and average power of given continuous-time and discrete-time signals and classify them as energy signals, power signals, or neither based on their energy and power characteristics.
Date of the Expt	
CO	CO1
GITHUB Link of the Expt	

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Calculate the energy and average power of given signals	Energy & Power Calculation	Correctly calculates signal energy and average power using appropriate mathematical expressions and MATLAB implementation independently.	Performs calculations correctly with minor numerical or coding errors.	Performs basic calculations but requires guidance or has noticeable errors.	Unable to correctly calculate energy or power.	20
Distinguish energy signals and power signals	Signal Classification	Correctly classifies signals as energy or power signals based on calculated values and theoretical conditions.	Correctly classifies signals with minor errors or limited justification.	Performs basic classification but requires assistance or provides incomplete justification.	Unable to classify the signal correctly.	20
Implement numerical energy/power computation using MATLAB	MATLAB Implementation & Numerical Analysis	Develops accurate MATLAB code, obtains reliable numerical values and appropriately selects the integration/summation interval.	Code and numerical results are mostly correct with minor errors.	Implements the basic computation with assistance or noticeable numerical errors.	Unable to implement the required MATLAB computation.	20
Compare analytical and simulated results	Theoretical Validation & Comparison	Accurately compares theoretical and simulated values, calculates deviation where applicable and explains discrepancies clearly.	Provides correct comparison with minor deviations and reasonable interpretation.	Provides basic comparison but validation or interpretation is limited.	Unable to compare or validate the results.	20
Interpret the physical meaning of	Engineering Interpretation & Result	Clearly explains the physical meaning of energy and power and	Provides correct interpretati	Provides basic interpretatio	Unable to explain	20

signal energy and power		gives a technically sound conclusion independently.	on with minor gaps.	n with guidance.	the meaning or significa nce of energy and power.	
Total						100

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Calculate the energy and average power of given signals		2		
Distinguish energy signals and power signals		2		
Implement numerical energy/power computation using MATLAB		2		
Compare analytical and simulated results		2		
Interpret the physical meaning of signal energy and power		2		
		Total		
		Course Faculty Sign with date		

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Verification of System Properties: Linearity, Time Invariance, Causality and Stability	
Expt / Prog No:	6
Name of the Expt / Prog	To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.
Date of the Expt	
CO	CO1, CO3
GITHUB Link of the Expt	

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Verify linearity and time invariance of CT and DT systems	Linearity & Time-Invariance Verification	Correctly applies homogeneity, additivity and time-shift tests; obtains accurate numerical errors and independently classifies the system.	Performs the tests correctly with minor coding, numerical or interpretation errors.	Performs basic verification but requires guidance or has noticeable errors in one or more tests.	Unable to correctly verify or interpret linearity/time invariance.	20
Analyze and classify causal and non-causal systems	Causality Analysis	Correctly determines causality using system/input-output relationships and supports the classification with appropriate plots or mathematical reasoning.	Correctly classifies most systems with minor errors or limited justification.	Performs basic causality analysis but requires assistance or provides incomplete justification.	Unable to determine or justify whether the system is causal or non-causal.	20
Analyze memoryless and dynamic system properties	Memoryless & Dynamic System Analysis	Correctly distinguishes memoryless and dynamic systems and clearly relates the classification to the dependence of output on present/past/future input values.	Classifies systems correctly with minor errors or limited explanation.	Identifies basic system behavior but requires guidance or provides partial reasoning.	Unable to correctly distinguish memoryless and dynamic systems.	20
Verify BIBO stability and classify stable and unstable systems	Stability Verification	Correctly applies the BIBO stability criterion, obtains accurate simulation results and clearly classifies stable and unstable systems.	Performs stability verification correctly with minor errors in calculation or interpretation.	Performs basic stability analysis but requires assistance or has noticeable errors.	Unable to verify or correctly classify system stability.	20
Interpret system properties and communicate engineering conclusions	Engineering Analysis, Record & Viva	Provides technically accurate interpretation of all system properties, complete record, clear conclusion and confidently answers	Maintains proper record and explains system properties with minor gaps; answers most	Record and interpretation are adequate but require guidance; answers basic questions.	Incomplete record, limited interpretation and unable to answer basic questions.	20

		viva questions.	viva questions correctly.			
TOTAL						**100**

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Verify linearity and time invariance of CT and DT systems		2		
Analyze and classify causal and non-causal systems		2		
Analyze memoryless and dynamic system properties		2		
Verify BIBO stability and classify stable and unstable systems		2		
Interpret system properties and communicate engineering conclusions		2		
Total				
Course Faculty Sign with date				

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Fourier Series Analysis of Periodic Signals	
Expt / Prog No:	7
Name of the Expt / Prog	To analyze Fourier series of continuous-time periodic signals and analyze their waveform characteristics.
Date of the Expt	
CO	CO2
GITHUB Link of the Expt	

Evaluation Criterion	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**
Experimental setup and Fourier-series formulation	Correctly defines periodic signal, T0, f0 and formulates the Fourier-series model with accurate parameters.	Setup and formulation are correct with minor parameter or notation errors.	Basic setup is completed but requires guidance or has noticeable errors.	Unable to establish the periodic signal or Fourier-series formulation.	20
Fourier coefficients and harmonic analysis	Accurately computes a0, an and bn; determines harmonic magnitude/phase and identifies dominant harmonics.	Computes coefficients and harmonics correctly with minor numerical or interpretation errors.	Obtains basic coefficients but needs guidance or has several calculation gaps.	Unable to compute or interpret Fourier coefficients/harmonics.	20
Amplitude/phase spectrum interpretation	Correctly plots and interprets amplitude and phase spectra and relates them to signal characteristics.	Plots spectra correctly with minor interpretation gaps.	Produces basic spectra but interpretation is incomplete.	Unable to generate or interpret the spectra.	20
Fourier-series reconstruction and validation	Accurately reconstructs the waveform, compares it with the original, and evaluates reconstruction error.	Reconstruction and comparison are correct with minor errors.	Reconstruction is attempted but comparison/error analysis is incomplete.	Unable to reconstruct or validate the waveform.	20
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents clear plots/results, maintains complete record and explains findings confidently in viva.	Executes correctly with minor coding, presentation or explanation gaps.	Requires assistance; record or interpretation is partially complete.	Unable to execute/record the experiment or explain the results.	20

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Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Experimental setup and Fourier-series formulation		2		
Fourier coefficients and harmonic analysis		2		
Amplitude/phase spectrum interpretation		2		
Fourier-series reconstruction and validation		2		
MATLAB execution, record and engineering interpretation		2		
		Total		
		Course Faculty Sign with date		

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Fourier Series Analysis of Periodic Signals	
Expt / Prog No:	8
Name of the Expt / Prog	**To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal.**
Date of the Expt	
CO	CO2
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Experimental setup and Fourier-series formulation	Correctly defines periodic signal, T0, f0 and formulates the Fourier-series model with accurate parameters.	Setup and formulation are correct with minor parameter or notation errors.	Basic setup is completed but requires guidance or has noticeable errors.	Unable to establish the periodic signal or Fourier-series formulation.	20
Fourier coefficients and harmonic analysis	Accurately computes a0, an and bn; determines harmonic magnitude/phase and identifies dominant harmonics.	Computes coefficients and harmonics correctly with minor numerical or interpretation errors.	Obtains basic coefficients but needs guidance or has several calculation gaps.	Unable to compute or interpret Fourier coefficients/harmonics.	20
Amplitude/phase spectrum interpretation	Correctly plots and interprets amplitude and phase spectra and relates them to signal characteristics.	Plots spectra correctly with minor interpretation gaps.	Produces basic spectra but interpretation is incomplete.	Unable to generate or interpret the spectra.	20
Fourier-series reconstruction and validation	Accurately reconstructs the waveform, compares it with the original, and evaluates reconstruction error.	Reconstruction and comparison are correct with minor errors.	Reconstruction is attempted but comparison/error analysis is incomplete.	Unable to reconstruct or validate the waveform.	20
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents clear plots/results, maintains complete record and explains findings confidently in viva.	Executes correctly with minor coding, presentation or explanation gaps.	Requires assistance; record or interpretation is partially complete.	Unable to execute/record the experiment or explain the results.	20

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Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Experimental setup and Fourier-series formulation		2		
Fourier coefficients and harmonic analysis		2		
Amplitude/phase spectrum interpretation		2		
Fourier-series reconstruction and validation		2		
MATLAB execution, record and engineering interpretation		2		
		Total		
		Course Faculty Sign with date		

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Parseval's Relation and Harmonic Spectrum Analysis	
Expt / Prog No:	9
Name of the Expt / Prog	To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.
Date of the Expt	
CO	CO2
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Periodic-signal setup and time-domain power	Correctly generates the periodic signal and accurately evaluates average power using time-domain integration.	Correct signal and power calculation with minor numerical errors.	Basic power calculation is completed with guidance or noticeable errors.	Unable to generate the signal or determine time-domain power.	20
Fourier coefficients and harmonic power	Accurately computes Fourier coefficients and determines DC and individual harmonic power contributions.	Computes coefficients and harmonic powers correctly with minor errors.	Obtains basic harmonic values but needs guidance in power calculation.	Unable to compute or interpret harmonic power.	20
Parseval verification	Accurately compares time-domain and frequency-domain power and determines numerical agreement/error.	Verification is correct with minor numerical or interpretation errors.	Performs basic verification but comparison is incomplete.	Unable to verify Parseval's relation.	20
Harmonic spectrum and power distribution	Correctly plots harmonic spectrum and interprets distribution of power among DC and harmonics.	Spectrum is correct with minor interpretation gaps.	Produces a basic spectrum but interpretation is limited.	Unable to generate or interpret the harmonic spectrum.	20

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Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents calculations/plots clearly, maintains complete record and explains power relationships in viva.	Good execution and record with minor gaps.	Requires assistance or has incomplete documentation/interpretation.	Unable to execute, document or explain the experiment.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Periodic-signal setup and time-domain power		**2**		
Fourier coefficients and harmonic power		**2**		
Parseval verification		**2**		
Harmonic spectrum and power distribution		**2**		
MATLAB execution, record and engineering interpretation		**2**		
		Total		
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Parseval's Relation and Harmonic Spectrum Analysis	
Expt / Prog No:	10
Name of the Expt / Prog	**To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.**
Date of the Expt	
CO	CO2
GITHUB Link of the Expt	

Evaluation Criterion	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**
Periodic-signal setup and time-domain power	Correctly generates the periodic signal and accurately evaluates average power using time-domain integration.	Correct signal and power calculation with minor numerical errors.	Basic power calculation is completed with guidance or noticeable errors.	Unable to generate the signal or determine time-domain power.	20
Fourier coefficients and harmonic power	Accurately computes Fourier coefficients and determines DC and individual harmonic power contributions.	Computes coefficients and harmonic powers correctly with minor errors.	Obtains basic harmonic values but needs guidance in power calculation.	Unable to compute or interpret harmonic power.	20
Parseval verification	Accurately compares time-domain and frequency-domain power and determines numerical agreement/error.	Verification is correct with minor numerical or interpretation errors.	Performs basic verification but comparison is incomplete.	Unable to verify Parseval's relation.	20
Harmonic spectrum and power distribution	Correctly plots harmonic spectrum and interprets distribution of power among DC and harmonics.	Spectrum is correct with minor interpretation gaps.	Produces a basic spectrum but interpretation is limited.	Unable to generate or interpret the harmonic spectrum.	20

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Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents calculations/plots clearly, maintains complete record and explains power relationships in viva.	Good execution and record with minor gaps.	Requires assistance or has incomplete documentation/interpretation.	Unable to execute, document or explain the experiment.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Periodic-signal setup and time-domain power		**2**		
Fourier coefficients and harmonic power		**2**		
Parseval verification		**2**		
Harmonic spectrum and power distribution		**2**		
MATLAB execution, record and engineering interpretation		**2**		
		Total		
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Continuous-Time Fourier Transform and Spectrum	
Expt / Prog No:	11
Name of the Expt / Prog	To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.
Date of the Expt	
CO	CO2, CO3
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
CTFT computation and signal representation	Correctly generates representative continuous-time signals and obtains accurate FFT-based CTFT approximations with appropriate sampling.	CTFT computation is correct with minor sampling or numerical issues.	Basic transform computation is obtained with guidance or noticeable errors.	Unable to obtain a meaningful CTFT approximation.	20
Magnitude and phase spectrum analysis	Accurately obtains magnitude/phase spectra and identifies dominant frequency components.	Spectra are correct with minor peak or phase interpretation errors.	Basic spectral plots are produced but interpretation is incomplete.	Unable to generate or interpret magnitude/phase spectra.	20
Fourier-transform property verification	Correctly verifies the time-shift property by comparing computed and theoretical magnitude and phase responses.	Property verification is correct with minor discrepancies or explanation gaps.	Attempts verification but requires guidance or incomplete comparison.	Unable to verify or explain the transform property.	20

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Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Bandwidth and spectral-component analysis	Correctly identifies spectral components, applies the specified threshold and determines occupied bandwidth with engineering interpretation.	Bandwidth and components are determined correctly with minor errors.	Basic spectral analysis is completed but threshold/bandwidth interpretation is incomplete.	Unable to identify components or determine bandwidth.	20
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents neat time/frequency plots, maintains complete record and explains communication significance in viva.	Good execution and presentation with minor gaps.	Requires assistance or has incomplete record/interpretation.	Unable to execute, document or explain the results.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
CTFT computation and signal representation		**2**		
Magnitude and phase spectrum analysis		**2**		
Fourier-transform property verification		**2**		
Bandwidth and spectral-component analysis		**2**		
MATLAB execution, record and engineering interpretation		**2**		
		Total		
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Continuous-Time Fourier Transform and Spectrum	
Expt / Prog No:	12
Name of the Expt / Prog	**Verification of Fourier Transform Properties and Frequency-Domain Characteristics**
Date of the Expt	
CO	CO2, CO4
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
CTFT computation and signal representation	Correctly generates representative continuous-time signals and obtains accurate FFT-based CTFT approximations with appropriate sampling.	CTFT computation is correct with minor sampling or numerical issues.	Basic transform computation is obtained with guidance or noticeable errors.	Unable to obtain a meaningful CTFT approximation.	20
Magnitude and phase spectrum analysis	Accurately obtains magnitude/phase spectra and identifies dominant frequency components.	Spectra are correct with minor peak or phase interpretation errors.	Basic spectral plots are produced but interpretation is incomplete.	Unable to generate or interpret magnitude/phase spectra.	20
Fourier-transform property verification	Correctly verifies the time-shift property by comparing computed and theoretical magnitude and phase responses.	Property verification is correct with minor discrepancies or explanation gaps.	Attempts verification but requires guidance or incomplete comparison.	Unable to verify or explain the transform property.	20

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Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Bandwidth and spectral-component analysis	Correctly identifies spectral components, applies the specified threshold and determines occupied bandwidth with engineering interpretation.	Bandwidth and components are determined correctly with minor errors.	Basic spectral analysis is completed but threshold/bandwidth interpretation is incomplete.	Unable to identify components or determine bandwidth.	20
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents neat time/frequency plots, maintains complete record and explains communication significance in viva.	Good execution and presentation with minor gaps.	Requires assistance or has incomplete record/interpretation.	Unable to execute, document or explain the results.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
CTFT computation and signal representation		**2**		
Magnitude and phase spectrum analysis		**2**		
Fourier-transform property verification		**2**		
Bandwidth and spectral-component analysis		**2**		
MATLAB execution, record and engineering interpretation		**2**		
		Total		
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Laplace Transform, Inverse Transform, ROC and Stability	
Expt / Prog No:	13
Name of the Expt / Prog	To determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.
Date of the Expt	
CO	CO2
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Laplace-transform computation	Correctly determines the Laplace transform of the given continuous-time signal and interprets the resulting expression.	Transform is obtained correctly with minor symbolic or interpretation errors.	Basic transform is obtained with guidance or noticeable errors.	Unable to determine or interpret the Laplace transform.	20
Inverse Laplace reconstruction	Accurately obtains the inverse transform, reconstructs the time-domain signal and validates the waveform.	Reconstruction is correct with minor calculation or plotting errors.	Basic inverse transform is obtained but validation is incomplete.	Unable to obtain or validate the reconstructed signal.	20
ROC and signal-sidedness analysis	Correctly identifies poles and the ROCs for right-sided, left-sided and two-sided signals and explains their significance.	ROC analysis is correct with minor explanation gaps.	Identifies basic ROCs but requires guidance or has incomplete reasoning.	Unable to determine or explain the ROC.	20

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Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Poles, zeros and stability verification	Correctly determines poles/zeros, compares pole locations with the unit circle and classifies stability for the stated ROC.	Stability verification is correct with minor numerical or interpretation errors.	Basic pole-zero analysis is performed but stability reasoning is incomplete.	Unable to determine poles/zeros or classify stability.	20
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents clear transform/pole-zero plots, maintains complete record and explains stability implications in viva.	Good execution and record with minor gaps.	Requires assistance or has incomplete documentation/interpretation.	Unable to execute, document or explain the experiment.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Laplace-transform computation		**2**		
Inverse Laplace reconstruction		**2**		
ROC and signal-sidedness analysis		**2**		
Poles, zeros and stability verification		**2**		
MATLAB execution, record and engineering interpretation		**2**		
		Total		
		Course Faculty Sign with date		

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Laplace Transform, Inverse Transform, ROC and Stability	
Expt / Prog No:	14
Name of the Expt / Prog	**To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.**
Date of the Expt	
CO	CO2
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Laplace-transform computation	Correctly determines the Laplace transform of the given continuous-time signal and interprets the resulting expression.	Transform is obtained correctly with minor symbolic or interpretation errors.	Basic transform is obtained with guidance or noticeable errors.	Unable to determine or interpret the Laplace transform.	20
Inverse Laplace reconstruction	Accurately obtains the inverse transform, reconstructs the time-domain signal and validates the waveform.	Reconstruction is correct with minor calculation or plotting errors.	Basic inverse transform is obtained but validation is incomplete.	Unable to obtain or validate the reconstructed signal.	20
ROC and signal-sidedness analysis	Correctly identifies poles and the ROCs for right-sided, left-sided and two-sided signals and explains their significance.	ROC analysis is correct with minor explanation gaps.	Identifies basic ROCs but requires guidance or has incomplete reasoning.	Unable to determine or explain the ROC.	20
Poles, zeros and stability verification	Correctly determines poles/zeros, compares pole locations with the unit circle and classifies stability for the stated ROC.	Stability verification is correct with minor numerical or interpretation errors.	Basic pole-zero analysis is performed but stability reasoning is incomplete.	Unable to determine poles/zeros or classify stability.	20

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Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
MATLAB execution, record and engineering interpretation	Executes MATLAB independently, presents clear transform/pole-zero plots, maintains complete record and explains stability implications in viva.	Good execution and record with minor gaps.	Requires assistance or has incomplete documentation/interpretation.	Unable to execute, document or explain the experiment.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Laplace-transform computation		**2**		
Inverse Laplace reconstruction		**2**		
ROC and signal-sidedness analysis		**2**		
Poles, zeros and stability verification		**2**		
MATLAB execution, record and engineering interpretation		**2**		
		Total		
Course Faculty Sign with date				

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CT LTI system	
Expt / Prog No:	15
Name of the Expt / Prog	To formulate a continuous-time LTI system from a differential equation, obtain its transfer function under zero initial conditions, simulate impulse and step responses, and relate the system poles to its transient behavior and stability.
Date of the Expt	
CO	CO3
GITHUB Link of the Expt	

Evaluation Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weight age
Differential-equation formulation	Correctly interprets the given differential equation, identifies system variables and coefficients, and formulates the LTI model independently.	Formulates the model correctly with minor notation or coefficient errors.	Identifies the main variables but needs guidance or makes noticeable formulation errors.	Unable to interpret the equation or formulate a suitable LTI model.	20
Transfer-function and state-space representation	Correctly obtains the transfer function under zero initial conditions and derives a valid state-space representation.	Obtains a valid representation with minor algebraic or matrix-entry errors.	Produces a partial model but has errors or incomplete representation.	Unable to derive a valid transfer-function or state-space model.	20
MATLAB implementation and response simulation	Implements the model independently in MATLAB and correctly simulates the specified input and system output.	Simulation is substantially correct with minor coding or setup errors.	Runs a partial simulation with assistance or noticeable coding errors.	Unable to implement or simulate the system correctly.	20
Response analysis and dynamic interpretation	Accurately interprets transient and steady-state behaviour, relates the response to system coefficients, and checks whether the output is bounded.	Interprets the response correctly with minor gaps in explanation.	Provides a basic interpretation, but transient/steady-state analysis is incomplete.	Unable to interpret the response or relate it to system behaviour.	20
Plots, verification, record and conclusion	Presents clearly labelled input/output plots, verifies expected behaviour, maintains a complete record, and states a technically justified conclusion.	Plots and record are clear with minor omissions or limited verification.	Plots, verification, record or conclusion are incomplete and require improvement.	Missing or unclear plots, verification, record or conclusion.	20

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Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Differential-equation formulation		2		
Transfer-function and state-space representation		2		
MATLAB implementation and response simulation		2		
Response analysis and dynamic interpretation		2		
Plots, verification, record and conclusion		2		
		Total		
		Course Faculty Sign with date		

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CT LTI system	
Expt / Prog No:	16
Name of the Expt / Prog	To distinguish the responses produced by initial stored energy and external excitation, obtain the impulse response of a continuous-time LTI system, and verify that the total response is the sum of the zero-input and zero-state responses, with the zero-state response also obtained through convolution.
Date of the Expt	
CO	CO3
GITHUB Link of the Expt	

Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System modelling and transfer function	Correct model and H(s)	Minor notation error	Partial model	Incorrect model	20
Response analysis	Correctly distinguishes all responses	Mostly correct	Partial distinction	Major confusion	20
MATLAB code and execution	Correct, clear, reproducible	Minor code issue	Runs with help	Does not run	20
Graphs and interpretation	All plots labelled and interpreted	Plots mostly correct	Incomplete plots	Missing/incorrect plots	15
Convolution verification	Correct convolution and comparison	Small numerical discrepancy explained	Partial verification	No valid verification	15
Record, observations and conclusion	Complete and technically sound	Minor omissions	Several omissions	Incomplete	10

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
System modelling and transfer function		2		
Response analysis		2		
MATLAB code and execution		2		
Graphs and interpretation		1.5		

Convolution verification		1.5		
Record, observations and conclusion		1		
		Total		
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Sampling Theorem Verification and Sampling-Frequency Selection	
Expt / Prog No:	17
Name of the Expt / Prog	Generate a continuous-time sinusoidal signal and sample it at sub-Nyquist, Nyquist and super-Nyquist rates.
Date of the Expt	
CO	CO4
GITHUB Link of the Expt	

Objective	Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
Verify sampling theorem, sampling frequency and reconstruction accuracy.	Sampling-rate selection and Nyquist condition	Selects a suitable sampling frequency and justifies it using the highest signal frequency and Nyquist criterion.	Selects and justifies the rate correctly with minor gaps.	Chooses a plausible rate but gives limited justification.	Chooses an unsuitable rate or cannot apply the criterion.	20
Verify sampling theorem, sampling frequency and reconstruction accuracy.	Sampling and reconstruction implementation	Correctly generates samples and reconstructs the signal using appropriate MATLAB methods and parameters.	Implementation is correct with minor parameter errors.	Generates samples but reconstruction or parameters need guidance.	Unable to implement sampling/reconstruction correctly.	20
Verify sampling theorem, sampling frequency and reconstruction accuracy.	Aliasing and spectral interpretation	Accurately identifies aliasing or its absence from sampled waveforms/spectra and explains the cause.	Identifies the main spectral effects with minor interpretation errors.	Recognizes basic effects but explanation is incomplete.	Misidentifies aliasing or cannot interpret the spectrum.	20
Verify sampling theorem, sampling frequency and reconstruction accuracy.	Theoretical and numerical verification	Verifies results against the sampling theorem and quantifies relevant error/deviation.	Provides a valid comparison with small omissions.	Provides a basic comparison with limited calculations.	Does not validate results against theory.	20
Verify sampling theorem, sampling frequency and reconstruction accuracy.	Engineering judgement, plots and conclusion	Uses clear labelled plots and gives a justified sampling-	Plots and conclusion are sound with minor gaps.	Plots or recommendation are incomplete.	Plots are unclear or conclusion is unsupported.	20

and reconstruction accuracy.		frequency recommendation based on results.				100
Total						

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Verify sampling theorem, sampling frequency and reconstruction accuracy.		2		
Verify sampling theorem, sampling frequency and reconstruction accuracy.		2		
Verify sampling theorem, sampling frequency and reconstruction accuracy.		2		
Verify sampling theorem, sampling frequency and reconstruction accuracy.		2		
Verify sampling theorem, sampling frequency and reconstruction accuracy.		2		
Verify sampling theorem, sampling frequency and reconstruction accuracy.		2		
		Total		
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Discrete-Time Fourier Transform and Frequency-Domain Analysis	
Expt / Prog No:	18
Name of the Expt / Prog	Compute the DTFT of finite-length discrete-time sequences.
Date of the Expt	
CO	CO4
GITHUB Link of the Expt	

Objective	Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
Compute and interpret the DTFT of discrete-time signals.	DTFT formulation and computation	Forms the correct DTFT expression and evaluates it accurately for the specified sequence.	Expression and computation are correct with minor errors.	Obtains a partial expression or needs guidance.	Unable to formulate or compute the DTFT.	20
Compute and interpret the DTFT of discrete-time signals.	Magnitude and phase spectrum generation	Generates correctly scaled, labelled magnitude and phase spectra over an appropriate frequency interval.	Plots are correct with minor scaling/label issues.	Plots basic spectrum but with omissions.	Plots are incorrect or missing.	20
Compute and interpret the DTFT of discrete-time signals.	Spectral feature interpretation	Correctly explains periodicity, symmetry, dominant components, and spectral nulls as applicable to the sequence.	Interprets main features with minor omissions.	Identifies only obvious features.	Misinterprets or cannot explain spectral features.	20
Compute and interpret the DTFT of discrete-time signals.	Analytical and numerical validation	Checks computed values against a theoretical DTFT or known limiting/symmetry property.	Validation is sound with minor gaps.	Provides a limited check with guidance.	No meaningful verification is provided.	20
Compute and interpret the DTFT of discrete-time signals.	MATLAB implementation and conclusion	Uses appropriate frequency sampling and presents a technically justified conclusion connecting sequence	Implementation and conclusion are mostly correct.	Basic implementation with limited interpretation.	Unable to produce reliable results or conclusion.	20

		properties to spectrum.				100
Total						

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Compute and interpret the DTFT of discrete-time signals.		2		
Compute and interpret the DTFT of discrete-time signals.		2		
Compute and interpret the DTFT of discrete-time signals.		2		
Compute and interpret the DTFT of discrete-time signals.		2		
Compute and interpret the DTFT of discrete-time signals.		2		
		Total		
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Z-Transform, ROC and Inverse Z-Transform of Discrete-Time Signals	
Expt / Prog No:	19
Name of the Expt / Prog	To compute the Z-transform of a selected discrete-time signal using MATLAB. .
Date of the Expt	
CO	CO5
GITHUB Link of the Expt	

Objective	Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
Compute Z-transform and inverse Z-transform.	Transform Computation & MATLAB Implementation	Correctly evaluates X(z), computes inverse transform, and executes MATLAB code independently.	Obtains correct transform results with minor coding or notation errors.	Obtains partial results with assistance or noticeable errors.	Unable to compute or implement the transforms.	25
Determine and interpret the Region of Convergence.	ROC Analysis	Correctly identifies ROC $\|z\| > 0.5$ for the right-sided sequence and explains its relation to the pole.	Identifies the ROC with minor explanation gaps.	Gives a partially correct ROC or limited justification.	Cannot identify or explain the ROC.	25
Verify sequence reconstruction and pole-zero locations.	Reconstruction & Pole-Zero Analysis	Original and reconstructed samples agree; pole/zero plot is correct and clearly labelled.	Plots are mostly correct with minor discrepancies or labelling errors.	Produces basic plots but verification or interpretation is incomplete.	Cannot verify reconstruction or identify pole-zero locations.	30
Interpret stability and document results.	Result Verification, Record & Viva	Correctly relates unit-circle inclusion to stability and records clear, evidence-based conclusions.	Provides correct stability interpretation with minor omissions.	Gives a basic conclusion but justification is limited.	Makes an incorrect stability claim or provides inadequate documentation .	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Compute Z-transform and inverse Z-transform.		2.5		
Determine and interpret the Region of Convergence.		2.5		
Verify sequence reconstruction and pole-zero locations2s		3		

Interpret stability and document results.		2		
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STFT and Wavelet-Based Time-Frequency Analysis of Non-Stationary Signals	
Expt / Prog No:	20
Name of the Expt / Prog	To generate a non-stationary signal containing time-varying frequency components and a transient. .
Date of the Expt	
CO	CO5
GITHUB Link of the Expt	

Objective	Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
Analyze non-stationary signals using STFT and wavelet methods.	Signal Generation & MATLAB Implementation	Generates the chirp and transient correctly, uses appropriate sampling and executes STFT/CWT code independently.	Generates the signal and runs both analyses with minor coding or parameter errors.	Produces a basic signal/analysis with assistance or noticeable setting errors.	Cannot generate the specified signal or execute the required analysis.	25
Identify time-varying frequency components and transient events.	STFT/Spectrogram Analysis	Correctly interprets the rising 50–300 Hz ridge and identifies transient energy near 150 Hz during 0.8–1.0 s.	Identifies the chirp and transient with minor interpretation gaps.	Recognizes some features but gives incomplete time/frequency interpretation.	Cannot identify the chirp trajectory or transient in the spectrogram.	25
Compare STFT and wavelet representations.	Wavelet/Scalogram Analysis	Produces a readable scalogram and accurately compares time-frequency localization and multiresolution behavior.	Produces and interprets the scalogram with minor errors.	Generates a basic scalogram but comparison is limited or needs guidance.	Cannot generate or interpret the wavelet scalogram.	30
Validate and document time-frequency results.	Result Verification & Documentation	Labels plots, confirms expected frequency/time regions, and records a clear evidence-based conclusion.	Plots and conclusion are mostly correct with minor omissions.	Documentation is incomplete or conclusion is weakly supported.	Results are unlabelled, inconsistent, or unsupported by evidence.	20

Objective / Assessment Criterion	Level	Max Weightage (10)	Mark Obtained	Mark (100)
Analyze non-stationary signals using STFT and wavelet methods.		2.5		
Identify time-varying frequency components and		2.5		

transient events.				
Compare STFT and wavelet representations.		3		
Validate and document time-frequency results.		2		
Total				
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SAVEETHA
SCHOOL OF ENGINEERING

DEPARTMENT OF ECE
LABORATORY INDEX

Course Code: ECA03
Course Title: Signals and Systems
Student Name: _____
Register Number: _____
SLOT: _____

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